Research Engineer or Postdoc: Open-sourcing and Privacy–Utility Evaluation of a Foundation Model for the French National Health Data System (SNDS)





Supervision:

- Judith Abécassis, Soda, Inria (judith.abecassis@inria.fr), researcher in AI for health
- Audrey Bergès, DREES, (audrey.berges@sante.gouv.fr), Deputy Head of the Innovation and Health Evaluation Office

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Position details Inria's SODA team (Statistics, Optimization, and Data Science for Health), in partnership with the French Directorate for Research, Studies, Evaluation and Statistics (Drees, Ministry of Health), is offering a 15 month position (**research engineer or postdoctoral researcher**, depending on the candidate's profile and experience) to open-source and evaluate the privacy–utility trade-off of **openSNDS2vec**, a foundation model trained on the healthcare trajectories of the entire French population, as part of the SeqNDS project.

We are looking for a candidate with a strong background in machine learning, statistics, or applied mathematics, with hands-on experience in data science and software engineering. Experience with privacy-preserving machine learning, differential privacy, or health data is a strong asset but not a strict requirement. Excellent Python skills are essential, together with good software engineering practices (testing, packaging, documentation, version control). A genuine interest in making advanced methods accessible to non-specialists, and in working with people from very different backgrounds such as statisticians, epidemiologists, legal experts, physicians, is central to this role. Fluency in written and spoken French and English is required.

The position will be split between Inria's SODA team at Inria Saclay and the Drees "Innovation and Evaluation in Health" Lab, 78 rue Olivier de Serres, Paris 15e. Applications (CV and cover letter) should be sent by email to Audrey Bergès and Judith Abécassis. The position is open until filled, with a target start date in October 2026.

Background The SNDS (Système national des données de santé) is one of the richest medico-administrative databases in the world, covering the healthcare trajectories of more than 70 million French residents over a 5-year depth of history. Since 2024, the SeqNDS project has applied state-of-the-art natural language processing methods (Word2Vec, BERT, GPT-style architectures) to these sequences of care, producing several foundation models inspired by Life2Vec [3], BEHRT [2], and Delphi-2M [4]. These models achieve strong predictive performance across mortality and 184 ICD-10-based diagnostic targets, and have been described in a recent Drees publication [1].

The next step is to share the resulting representations (*ie*,vector embeddings for each medical code (ICD-10, ATC, CCAM, NABM, CIP, etc.)) as an open-source foundation model, **openSNDS2vec**, so that any team working on health data structured around standard nomenclatures (including hospital data warehouses) can benefit from knowledge learned on the exhaustive French population, without needing direct access to individual-level SNDS data. Because these representations are derived from highly sensitive health data, a rigorous, quantified privacy audit is required before any public release.

Missions

1. **Open-source packaging and code sharing.** Prepare, document, and release the SeqNDS codebase — including *openSNDS2vec* and the companion data-preparation tool *SNDS2table* — as clean, tested, reusable open-source software (Apache-2 license), suitable for reuse by teams without deep learning or AI expertise,

both on Drees infrastructure and on the French Health Data Hub Platform (Plateforme des données de santé, PDS), or the CNAM platform.

2. **Privacy–utility trade-off of the shared model.** Design and implement a k-anonymization procedure for the co-occurrence matrix underlying openSNDS2vec, so that a co-occurrence between medical codes is only released if it appears in the trajectories of at least k distinct individuals. Systematically characterize the trade-off between the choice of k, the risk of re-identification, and the retained predictive utility of the embeddings (measured via the AUC on the mortality and diagnosis-prediction benchmarks already established in SeqNDS).

3. **Privacy attacks and robustness evaluation using CNIL's PANAME framework.** Apply and adapt privacy attacks (membership inference) using the **PANAME** framework developed with Inria's Premedical team and the CNIL, to empirically evaluate the residual re-identification risk of the de-identified representations before any public release, in direct collaboration with the Premedical team.

4. **Pedagogical example gallery for epidemiologists.** Design and write a gallery of concrete, well-documented usage examples (in the spirit of scikit-learn's example gallery), in R and Python, showing how to query and reuse the vector representations in classical epidemiological workflows (e.g. enriching a small hospital cohort, building predictive models, exploring patient trajectory clusters). This work will be carried out in close collaboration with the CEPHEPI epidemiology network to ensure the material is accessible to non-AI-specialist researchers and clinicians.

5. **Documentation and dissemination.** Contribute to the scientific write-up of the de-identification methodology (submission to a peer-reviewed journal), and to presentations to Drees leadership, Inria teams, the CNIL, and the broader SNDS user community (e.g. PDS meetups, EMOIS, EPICLIN).

Profile sought

- PhD (for a postdoctoral position) or engineering/master's degree with significant applied experience (for a research-engineer position), in machine learning, statistics, applied mathematics, or a related field;
- Strong, demonstrable coding skills in Python, standard scientific libraries (scikit-learn, pandas), and good software-engineering practices;
- Genuine interest or willingness to build expertise in privacy-preserving machine learning and differential privacy; prior exposure to re-identification attacks or privacy audits is a plus;
- Strong pedagogical skills and enjoyment of writing clear, didactic documentation and examples for non-specialist audiences;
- Ease and enthusiasm for working with people from very different backgrounds (statisticians, epidemiologists, physicians, legal and regulatory experts);
- Good written and oral communication skills in French and English;
- Experience with health data (SNDS, PMSI, EHR, or similar) is appreciated but not required.

Environment The successful candidate will join Inria's Soda team, led by Gaël Varoquaux. Soda is doing computational and statistical research, both fundamental and applied, to harness large databases on health and society, and is known for developing scikit-learn (the third most downloaded machine learning library worldwide), skrub, and the tabular foundation model TabICL.

Part of the work will take place within the Drees **"Innovation and Evaluation in Health" Lab**, a multidisciplinary team of data scientists, data engineers, and statisticians, at 78 rue Olivier de Serres, Paris 15^{e}. The candidate will also interact with the CEPHEPI epidemiology team (AP-HP) for external validation of the shared representations, and with the CNIL in the context of the project's privacy-by-design review process.

The salary range is EUR 2,234–3,090 net per month, depending on experience (Inria salary grid), i.e. EUR 32,299–44,675 gross annually. The successful candidate will benefit from a standard French employment package, including partial reimbursement of commuting costs, 7 weeks of paid annual leave, RTT days, and comprehensive social security coverage.

References

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